

PABLO WILLIAMS

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WORK EXPERIENCE

Data Analyst, APJ Kapital - London, UK	Sep 2023 - Present
- Analyse company filings (10-Ks, 10-Qs, 8-Ks), earnings-call transcripts, and real-time market news to produce structured investment signals and buy/hold/sell recommendations for a portfolio of over 30 equities. - Build and maintain SQL data pipelines in PostgreSQL and Tableau dashboards that track portfolio P&L, sector allocation, and factor exposures for weekly portfolio review meetings. - Designed an n8n automation workflow that ingests daily market-update videos via Discord, transcribes them with Whisper, summarises with GPT-4o, and distributes briefs to Telegram and email.	
AI Investment Programming Intern, MDOTM - Milan, Italy	May 2022 - Oct 2022
- Built AI-driven investment tools and customer-segmentation models in Python and R, applying preprocessing, feature engineering, and supervised learning to sharpen predictive accuracy and targeting. - Worked with a 40-person cross-functional team to embed ML into investment workflows, running data-quality reviews and turning stakeholder feedback into measurable performance gains. - Played a key role in MDOTM's ML platform, which won the 2022 AIFIN Financial Innovation Award and was named Best Product of the Year at the New York FinTech Awards.	

PROJECTS

Calibration-Optimised Equity Outperformance Python, XGBoost, LightGBM, Optuna, SHAP	Jan 2026
- Tested whether ECE-optimised classifiers deliver better portfolio returns under half-Kelly sizing across 457 S&P 500 stocks (2010-2024). Random Forest achieved the best calibration (ECE = 0.026), a Sharpe of 0.77, and 68.1% total return.	
Churn Propensity Modelling Python, XGBoost, Random Forest, Scikit-learn	Mar 2026
- End-to-end churn prediction pipeline on IBM Telco data (7,043 customers). XGBoost achieved AUC-ROC 0.847; top 20% of risk-ranked customers capture 55% of all churners. Includes lift/gain analysis, calibration curves, and decile-level targeting.	
Online Community Topic Discovery Python, Gensim, LDA, NMF, NLTK, pyLDAvis	Mar 2026
- Unsupervised NLP pipeline comparing LDA and NMF on 14,000+ discussion posts. Applied urgency scoring, pain point extraction, extractive summarisation, severity clustering, and topic-category cross-analysis to each discovered topic.	
Polish Corporate Bankruptcy Prediction Python, LightGBM, XGBoost, Optuna, SHAP	Dec 2025
- Leak-proof ML pipeline on 10,503 company-year records with 64 financial ratios, handling 50% missingness and 20:1 class imbalance. LightGBM reached AUPRC of 0.6249 (13.3x over baseline) and AUC-ROC of 0.926.	
RiskTerrain Python, LangGraph, SurrealDB, Docker	Mar 2026
- 6-node LangGraph pipeline that ingests live geopolitical events, runs multi-hop supply-chain traversal across 154 S&P 500 companies, and scores cascading risk in under 8 seconds. Built at the LangChain x SurrealDB London Hackathon.	
AlphaQuant Factor Timing Python, XGBoost, scikit-learn, cvxpy	Sep 2024
- ML-enhanced factor timing model blending Value, Momentum, Quality, and Low Volatility with XGBoost regime detection. Backtested Sharpe of 0.94 (47% above S&P 500) and maximum drawdown cut by 46%.	
Sentinel Credit Risk Prediction Python, XGBoost, LightGBM, SHAP	Dec 2024
- Ensemble gradient boosting for consumer loan default prediction on 15,000 Lending Club records. AUC-ROC of 0.89 with Platt-scaled probabilities and SHAP-based explainability dashboards.	
Vista Zero-Shot Fashion Detection Python, OWL-ViT, CLIP, YOLOv8	Jan 2024
- Zero-shot detection using CLIP and OWL-ViT with knowledge distillation into YOLOv8-nano. Hit 88.7% of supervised performance (mAP@0.5: 0.751) with 99% annotation cost reduction across 44,441 images.	
Vertex SaaS Churn Prediction Python, XGBoost, Survival Analysis	Feb 2024
- B2B churn prediction on 20,000 customer records reaching AUC-ROC of 0.91 and precision of 0.85, flagging at-risk customers through engagement metrics and behavioural patterns.	
Project Lockdown Python, VRP Optimisation, Google Maps API	Jul 2024
- Emergency food distribution system for 52,000 residents over 28 days, using exponential smoothing for demand forecasting and vehicle routing optimisation with capacity constraints.	

EDUCATION

MSc Business Analytics - UCL	Sep 2025 - Sep 2026
Predicted: Distinction (72%) Modules: NLP, Machine Learning for Domain Specialists, Predictive Analytics, Data Visualisation, Statistical Methods, Mathematical Analysis.	
BSc Business Analytics (Minor: Accounting & Finance) - Lancaster University	Sep 2021 - Jul 2025
Achieved: Upper Second-Class Honours (2:1). Minor: Upper Second. Modules: Data Mining, Data Visualisation, Spreadsheet Modelling, Statistical Analytics.	

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL (PostgreSQL, DuckDB), R (tidyverse, ggplot2), JavaScript
ML: XGBoost, LightGBM, Random Forest, PyTorch, TensorFlow, LLMs (LangChain, Ollama), Optuna, SHAP
Data Engineering: PySpark, MongoDB, DuckDB, Parquet, Docker, n8n, AWS (S3)
Visualisation & BI: Tableau, Streamlit, Matplotlib, Seaborn, LaTeX | NLP: Sentence-Transformers, RAG, Whisper, GPT-4o

CERTIFICATIONS

Bloomberg: Finance Fundamentals, Market Concepts, Spreadsheet Analysis | **Google:** Data Analytics, IT Automation with Python
IBM: BI Analyst, Applied Software Engineering, AI Enterprise Workflow | **Yale:** Financial Markets | **LBS:** Financial Analysis | **Duke:** MLOps

LEADERSHIP, LANGUAGES & INTERESTS

Mathematics & CS Tutor GCSE, IB, SAT, GMAT levels; tracking student progress with Excel and Tableau diagnostics.	Feb 2024 - Present
Data Science Society Member Assisted peers with ML and analytics projects weekly.	Oct 2022 - May 2025
Volunteer, Milan Dog Kennel Walking and caring for rescue dogs every Sunday.	Sep 2015 - Sep 2021
Languages: Italian (Native) English (Native) Spanish (Fluent) German (Beginner)	